CAPSTONE PROJECT REPORT

Advanced Technical Analysis Assignment

JUNE 24, 2025

ABHIMANYU BABBAR

TABLE OF CONTENTS

S.No	Topic	Page No.
1.	**Introduction**	**2-3**
1.1.	Objective of the Capstone Project	2-2
1.2.	Scope and Relevance of Technical Analysis in Financial Markets	2-2
1.3.	Methodology Used	2-2
1.4.	Tools and Data Sources	3-3
2.	**Understanding Advanced Technical Indicators**	**4-8**
2.1.	Relative Strength Index (RSI)	4-4
2.2.	Moving Averages (SMA, EMA)	5-5
2.3.	MACD – Signal Line & Histogram	5-5
2.4.	Bollinger Bands – Rage Bound Strategy	6-6
2.5.	Fibonacci Retracement	6-6
2.6.	On Balance Volume (OBV)	7-7
2.7.	Volume Analysis – High Volume Stock Selection	7-7
2.8.	Price Gaps in Candlesticks	8-8
3.	**Chart Patterns and Candlestick Formations**	**9-16**
3.1.	Reversal Chart Patterns	9-10
3.2.	Continuation Chart Patterns	11-12
3.3.	Candlestick Patterns (Reversal and Confirmation)	12-15
3.4.	Gap Analysis in Candlestick Charts	16-16
4.	**Technical Analysis Theories in Action**	**17-20**
4.1.	Dow Theory – Trend Structure & Market Phases	17-17
4.2.	Eliott Wave Theory – Market moves in Waves	18-18
4.3.	Fibonacci Retracement – Measuring Probable Pullbacks	19-19
4.4.	Efficient Market Hypothesis – Challenge to Technical Analysis	19-20
4.5.	Random Walk Theory - Price is Unpredictable	20-20
5.	**Derivatives Market Integration**	**21-23**
5.1.	Overview of Derivative Instruments Used	21-21
5.2.	Open Interest (OI) Analysis	21-21
5.3.	Option Chain Analysis: Support and Resistance Zones	21-22
5.4.	Option Greeks Used in Decision Making	22-22
5.5.	Combining Technical with Derivatives for Trade	22-22
5.6.	Nifty 50 and Nifty 200 Focus	23-23
6.	**Trade Summary Table**	**24-25**
7.	**Elaborate Analysis of the Trades**	**26-35**
8.	**Reflection Report**	**36-38**

1. INTRODUCTION

1.1. Objective of the Capstone Project

This Capstone Project sets out to use technical analysis tools, chart patterns, and derivative market indicators to make informed decisions when trading in the equity market. Using high-probability trade setups in Nifty 50/Nifty 200 stocks, the project hopes to link trading theory learned in an academic setting and its application in the real world. The study focuses on obtaining a better understanding of how technical indicator, derivative indicator and market timing could improve risk management, alongside identifying trend direction.

1.2. Scope and Relevance of Technical Analysis in Financial Markets

In the today's active financial market environment, with swings in price due to the level of investor sentiment, surges in volume, and news events, technical analysis gives you a structure to take a data driven approach to your trading. Unlike the fundamental analysis route where you are considering the intrinsic value of what you're trading, technical analysis is complementary to more of a human side, the market psychology, the patterns of price action, and the momentum in the other direction.

This study has relevance for individual or retail traders, as well as for institutional professionals in a variety of roles, including equity research, proprietary trading, and portfolio management. The various techniques covered in the project, including RSI, MACD, Bollinger bands, Fibonacci retracements, and candlestick gap analysis, can be applied and have practical use in the approaches of market participants determining price action and taking trading action.

1.3. Methodology Used

The study utilizes the following:

a. Chart Based analysis using TradingView.
b. Leading and Lagging Indicators like RSI, Bollinger Bands, MACD, EMA's.
c. Volume and Open Interest Analysis from the NSE Option Chain
d. Derivatives information, including interpretation of Option Greeks (Delta, Theta, Gamma, Vega).
e. Pattern recognition, along with both continuation and reversal patterns displayed with annotated chart drawings.

Trade decisions were based on confluences of indicators, confirmation patterns, trading volume behaviour, and open interest patterns. Trades were primarily conducted in high-volume stocks from Nifty 50 and Nifty 200 to ensure liquidity and tighter spreads.

1.4. Tools and Data Sources

The various tools and platforms used throughout the project are as follows:

a. TradingView: For charting, drawing patterns, tracking indicators, identifying breakouts and Fibonacci setups.
b. NSE India Option Chain: For live Futures and Options Data, and Open Interest data.
c. Mock Broker Terminals: FrontPage for live price entries, exits, targets and stop losses.
d. Excel Sheets: For tracking each trade, along with its entry, exit, target and stoploss.

Through collaboration of the above tools along with the combination of theory and real trades, this project seeks to present how technical analysis with the application of risk management and discipline can assist traders in improving their decision making in terms of trading and ultimately maintaining the risk-reward ratio.

2. UNDERSTANDING ADVANCED TECHNICAL INDICATORS

This section covers the quantitative and visual tools used in technical analysis that helps traders make informed decisions. These are classified into leading (predictive) or lagging (confirmatory) indicators:

2.1. Relative Strength Index (RSI)

a. Type: Leading Indicator

b. Formula:

$$RSI_{\text{step one}} = 100 - \left[\frac{100}{1+\frac{\text{Average gain}}{\text{Average loss}}}\right]$$

$$RSI_{\text{step two}} = 100 - \left[\frac{100}{1+\frac{(\text{Previous Average Gain}\times 13)+\text{Current Gain}}{((\text{Previous Average Loss}\times 13)+\text{Current Loss})}}\right]$$

c. Interpretation:

- RSI > 70: Overbought (Possible Reversal or Pause)
- RSI < 30: Oversold (Possible Bounce or Reversal)

d. Use in Trading:

- Identify Divergence (Price makes new high, RSI does not), Potential Reversal indication.
- Combine with Candlestick confirmation near the support/resistance levels.

e. Chart Inclusion:

- RSI indicator below the price chart with annotations.

Figure 2.1. Candlestick charts along with RSI

2.2. Moving Averages (SMA, EMA)

a. Type: Lagging Indicator
b. Formula:
    - SMA Formula: Average of closing prices over n days.
    - EMA Formula: EMA = [Closing Price * K] + [Previous EMA * (1-K)]
    - K = 2/(n+1)
c. Common Periods: 20 EMA (Short Term), 50 EMA (Mid Term), 200 EMA (Long Term).
d. Usage:
    - Trend Confirmation
    - Dynamic Support/Resistance
    - Crossovers (Eg: Golden Cross – Bullish; Death Cross – Bearish)
e. Chart Inclusion:
    - Plot SMA/EMA Lines, mark crossover entries.

Figure 2.2. EMA vs SMA

2.3. MACD – Signal Line & Histogram

a. Type: Lagging Indicator
b. Formula:
    - MACD Line = 12 EMA – 26 EMA
    - Signal Line = 9 EMA of MACD Line
c. Interpretation:
    - MACD > Signal; High probability in Bullish Momentum
    - MACD < Singal; High probability in Bearish Momentum
    - Zero Line Crossovers; High probability in trend confirmation.
d. Use in trading:
    - Combine with RSI for trade confirmation.
e. Chart Inclusion:
    - MACD Histogram, Line chart with crossover points.

Figure 2.3. MACD and Signal Line

2.4. Bollinger Bands – Range Bound Strategy

a. Type: Volatility and Trend Indicator

b. Formula:

- Upper Band = SMA + (2*Standard Deviation)
- Middle Band = SMA
- Lower Band = SMA – (2*Standard Deviation)

c. Interpretation:

- Price touching upper band indicates that the stock is overbought.
- Price touching lower band indicates that the stock is oversold.
- Squeeze indicates breakout is to be expected.

d. Use in Trading:

- Range-bound strategies
- Bollinger bands + RSI for Confirmation

e. Chart Inclusion:

- Show Bollinger Bands with squeeze pattern marked.\

Figure 2.4. Bollinger Bands – Upper Band, Middle Band and Lower Band

2.5. Fibonacci Retracement

a. Type: Leading Indicator / Support-Resistance Tool

b. Key Levels: 23.6%, 38.2%, 50%, 61.8%, 78.6%

c. Use in Trading:

- Entry after retracement in uptrend.
- Target or stop-loss placement.

d. Chart Inclusion:

- Annotate Fibonacci tool on swing high to swing low.

Figure 2.5. Fibonacci Retracement

2.6. On Balance Volume (OBV)

a. Type: Leading Volume Indicator

b. Formula:

- If Today's price > Yesterday's Close, OBV = OBV + Volume
- If Today's Close > Yesterdays Price, OBV = OBV - Volume

c. Use in Trading:

- Confirms price trends.
- Divergence signals trend change.

d. Chart Inclusion:

- OBV plotted below chart with trendlines.

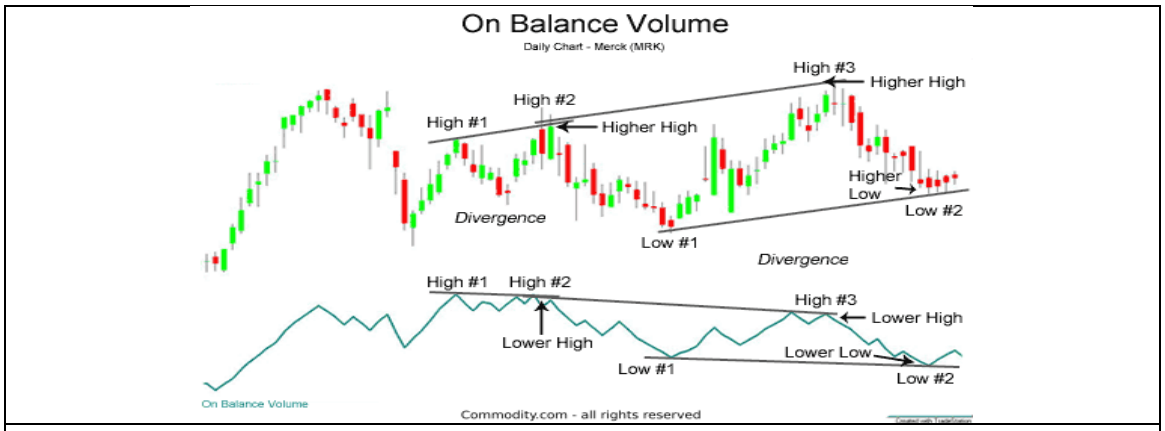

Figure 2.6. On Balance Volume (OBV) Indicator

2.7. Volume Analysis – High Volume Stock Selection

a. Use in Trading:

- Confirm breakout or breakdown.
- Spot accumulation/distribution.

b. Tips:

- Breakout with volume = Real Move
- Breakout without volume = False breakout risk

c. Chart Inclusion:

- Candle chart with volume bars, note divergence or breakout points.

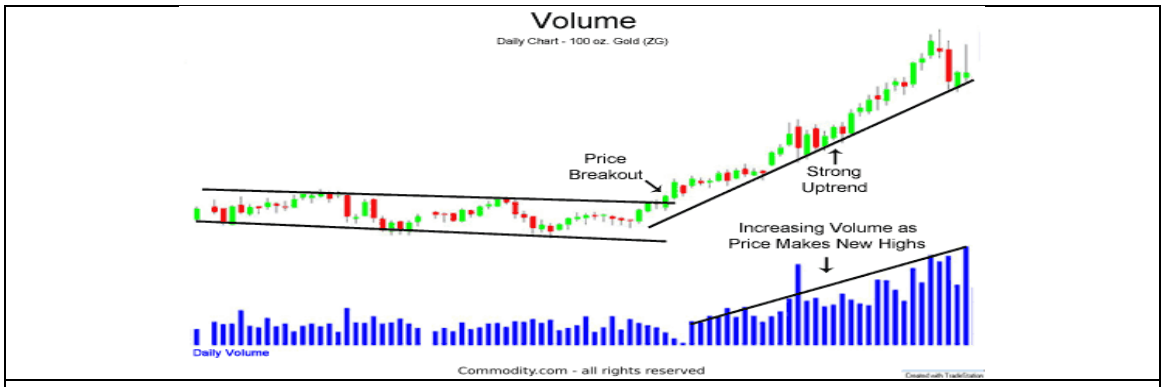

Figure 2.7. Volume Analysis for a daily chart

2.8. Price Gaps in Candlesticks

a. Types of Gaps:
   - Common Gap
   - Breakaway Gap
   - Runaway Gap
   - Exhaustion Gap

b. Use in Trading:
   - Trade continuation after breakaway gap.
   - Book profits after exhaustion gap.

c. Chart Inclusion:
   - Gap zones marked with entry and stop-loss.

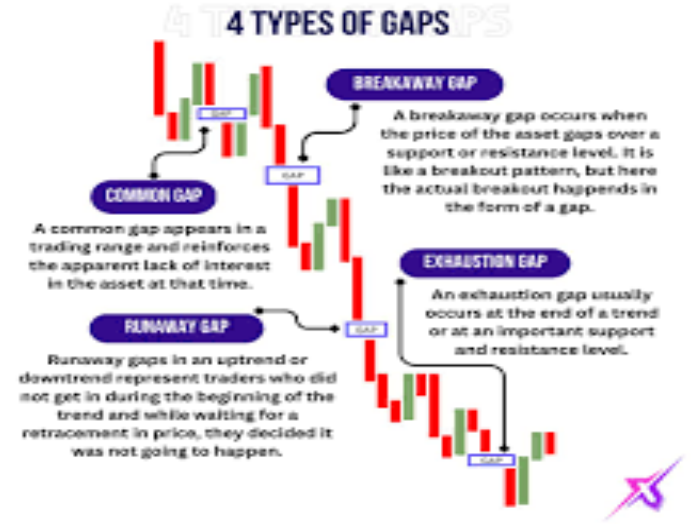

Figure 2.8. Price Gaps noted in Candlesticks

3. CHART PATTERNS AND CANDLESTICK FORMATIONS

Chart patterns form the backbone of price-based trading. When supported by volume and momentum indicators, they provide high probability entry and exit zones. This section classifies patterns into reversal. Continuous, candlestick-based, and gap related patterns with actionable trade strategies.

3.1. Reversal Chart Patterns

These patterns signal a possible change in the trend direction. Used primarily at tops or bottoms.

a. Double Top & Double Bottom

- Double Top: Bearish reversal pattern formed after an uptrend with two consecutive peaks at the same level.
- Double Bottom: Bullish reversal pattern formed after a downtrend with two consecutive troughs at the same level.
- Confirmation: Neckline breakout on volume is essential; a close below (Double Top) or above (Double Bottom) the neckline increase the probability of the move.
- Strategy:
  - ❖ Entry: On candle close below neckline (Double Top) or above neckline (Double Bottom).
  - ❖ Stop-Loss: Just above recent peak (Double Top) or below recent trough (Double Bottom).
  - ❖ Target: Height from neckline to top/bottom projected in breakout direction.
  - ❖ Volume: Should Confirm the breakout direction.

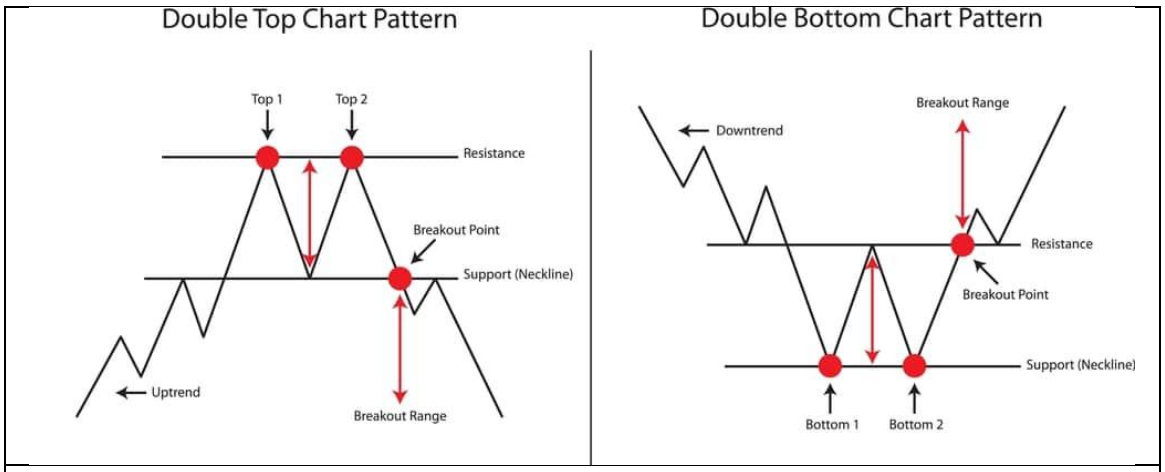

Figure 3.1.a. Reversal Chart Pattern of Double Top and Double Bottom

b. Head and Shoulders / Inverse Head and Shoulders

- Head and Shoulders: Indicates a reversal from an uptrend to downtrend.
- Inverse Head and Shoulders: Reversal from downtrend to uptrend.
- Confirmation: Break of neckline with strong candle and volume.
- Strategy:
  - ❖ Entry: On close beyond neckline.
  - ❖ Stop-Loss: Just above (H&S) or below (Inverse H&S) the right shoulder.
  - ❖ Target: Distance from head to neckline projected from breakout point.
  - ❖ Volume: Right shoulder often forms on lower volume than head.

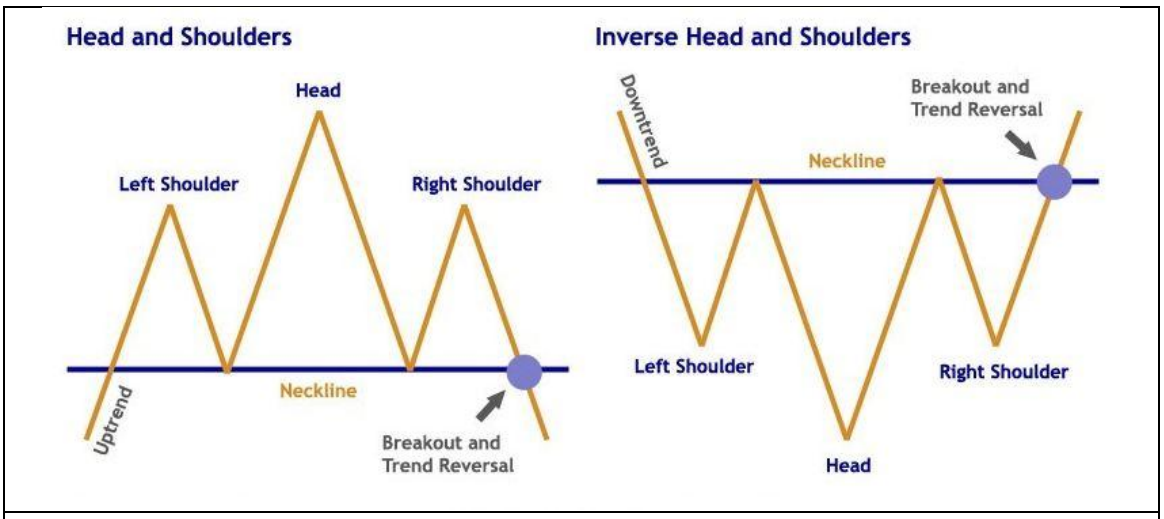

Figure 3.1.b. Reversal Chart Patterns of Head and Shoulders and Reverse Head and Shoulders

c. Cup and Handle

- Rounded base (Cup) followed by a slight downward channel (Handle) that breaks out upward.
- Confirmation: Breakout from the handle on volume surge.
- Strategy:
  - ❖ Entry: On breakout candle from handle
  - ❖ Stop-Loss: Just below the low of handle
  - ❖ Target: Height of cup projected upward from breakout point
  - ❖ Volume: Right shoulder often forms on lower volume than head.

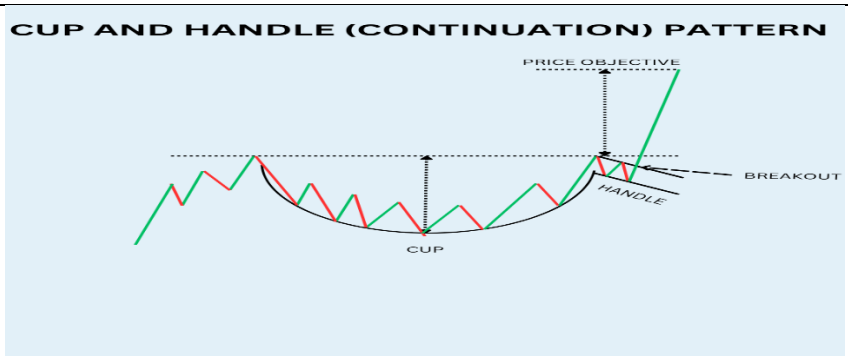

Figure 3.1.c. Reversal Chart Patterns of Cup and Handle

3.2. Continuous Chart Patterns

a. Falling and Rising Wedges

- Falling Wedges (Bullish): Downward sloping narrowing range; signals potential bullish breakout.
- Rising Wedges (Bearish): Upward sloping narrowing range; signals bearish breakout.
- Confirmation: Breakout of wedge line on volume; often preceded by declining volume during pattern formation.
- Strategy:
  - ❖ Entry: On breakout candle closing outside the wedge
  - ❖ Stop-Loss: Below/above the opposite edge of wedge pattern
  - ❖ Target: Height of wedge at its widest projected from breakout.
  - ❖ Exit: Exit if price loses momentum mid-way (watch RSI divergence).

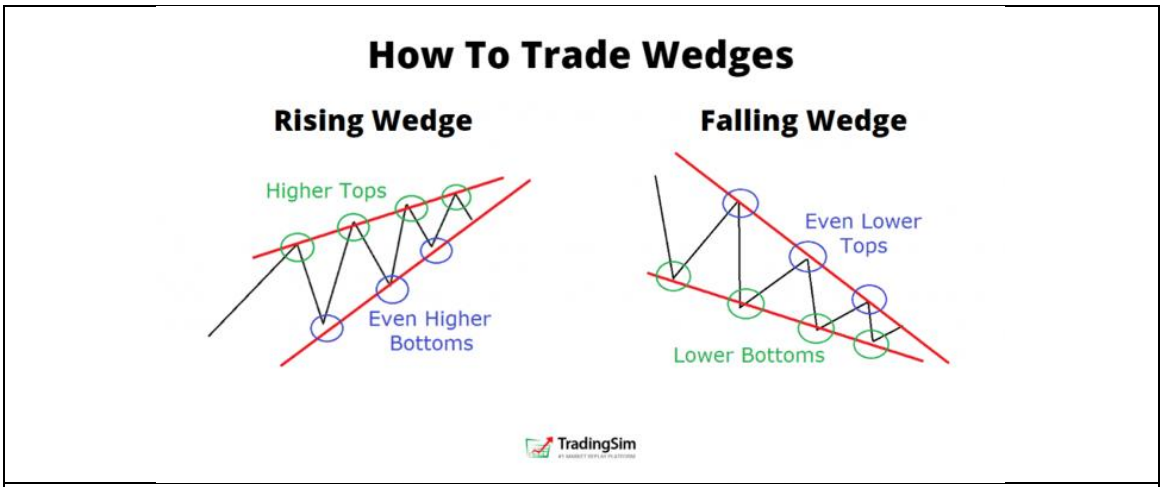

Figure 3.2.a. Continuous Chart Patterns of Falling and Rising Wedges

b. Bullish and Bearish Rectangle

- Bullish Rectangle: Horizontal consolidation in uptrend; breakout expected on upside.
- Bearish Rectangle: Range-bound price in a downtrend; breakdown expected downward.
- Confirmation: Multiple touches on support and resistance zones with a breakout candle and volume spike.
- Strategy:
  - ❖ Entry: On breakout above resistance (bullish) or breakdown below support (bearish).
  - ❖ Stop-Loss: Below support (bullish) or above resistance (bearish).
  - ❖ Target: Height of rectangle projected from breakout.
  - ❖ Exit: Near target or on signs of trend exhaustion (e.g., reversal candle).

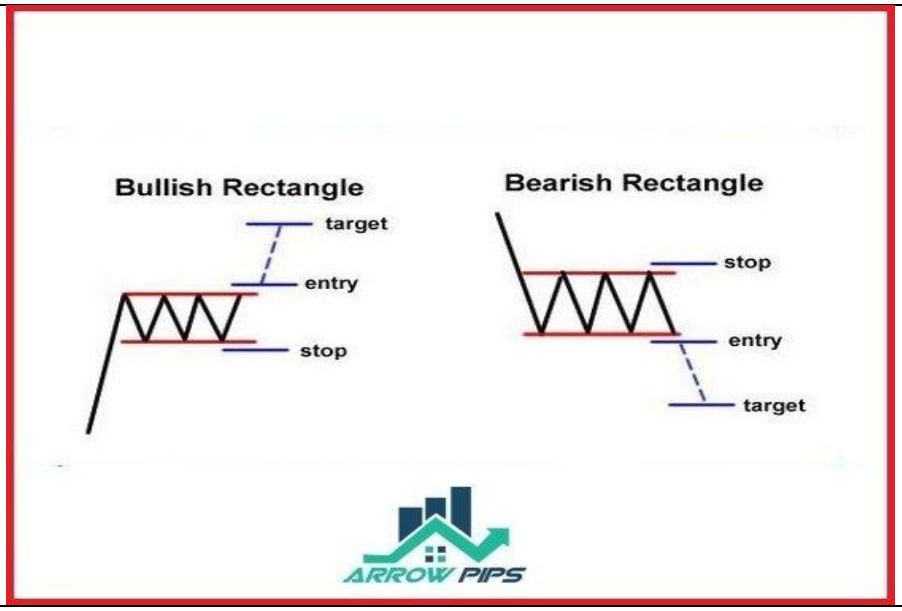

Figure 3.2.b. Continuous Chart Patterns of Bullish and Bearish Rectangle

3.3. Candlestick Patterns (Reversal and Confirmation)

a. Doji

- Indicates indecision; potential reversal depending on the context.
- Strategy:
  - ❖ Entry: After confirmation candle following Doji.
  - ❖ Stop Loss: Below/Above Doji's Wick
  - ❖ Target: Nearby support/resistance
  - ❖ Exit: If confirmation candle fails to follow through

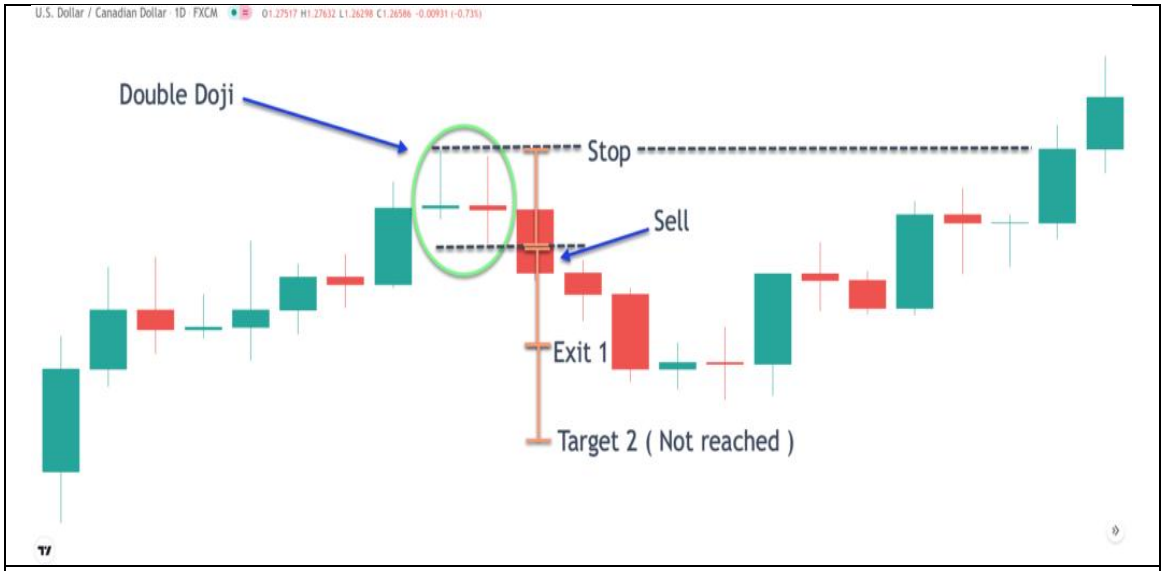

Figure 3.3.a. Doji Candlestick Chart Patterns

b. Marubozu

- Full-bodied candle without wicks, strong bullish (Green) or bearish (red) sentiment.
- Strategy:
  - ❖ Entry: On close of Marubozu or next candle in direction of trend.
  - ❖ Stop Loss: At midpoint or opposite end of the Marubozu
  - ❖ Target: Nearest resistance/ support level.
  - ❖ Exit: Trail SL aggressively due to fast move probability

Fig. Bullish Marubozu

Figure 3.3.b. Marubozu Candlestick Chart Pattern

c. Shooting Star

- Bearish reversal at top with long upper wick and small real body.
- Strategy:
  - ❖ Entry: After red confirmation candle below low of shooting star
  - ❖ Stop Loss: Above high of the wick
  - ❖ Target: Next support or 1:2 risk-reward level
  - ❖ Exit: If next candle forms higher high

Figure 3.3.c. Shooting Star Candlestick Chart Pattern

d. Morning Star

- Bullish 3-candle reversal at bottom: Red is Doji/Small is Green.
- Strategy:
  - ❖ Entry: On close of third green candle.
  - ❖ Stop Loss: Below Doji/Small candle low
  - ❖ Target: Resistance zone or previous swing high
  - ❖ Exit: Partial f price slows, rest on RSI near 70

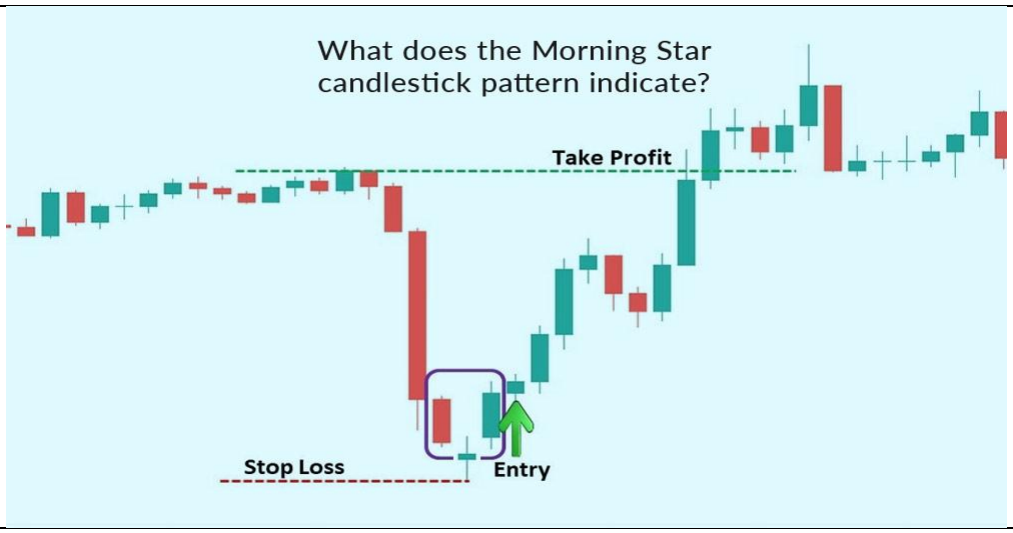

Figure 3.3.d. Morning Star Candlestick Chart Pattern

e. Bullish/ Bearish Engulfing

- Bullish Engulfing: Green Candle fully engulfs prior red candle
- Bearish Engulfing: Red Candle fully engulfs prior green candle
- Strategy:
  - ❖ Entry: After engulfing confirmation on volume
  - ❖ Stop Loss: Below/Above engulfing candle
  - ❖ Target: Based on prior support/resistance
  - ❖ Exit: When price hits overbought/oversold zones

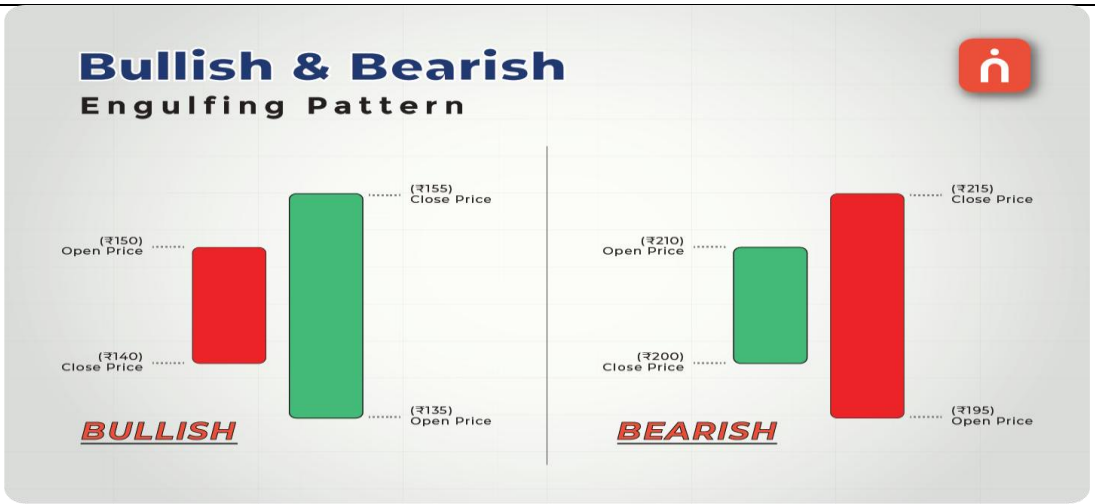

Figure 3.3.e. Bullish/ Bearish Engulfing Candlestick Chart Pattern

f. Hammer/ Inverted Hammer

- Hammer (Bullish): Long lower wick, small body at top
- Inverted Hammer (Bullish) Long Upper wick after downtrend
- Strategy:
    - ❖ Entry: After next green candle closes above high
    - ❖ Stop Loss: Below Wick Low
    - ❖ Target: Swing high or EMA crossover
    - ❖ Exit: If follow-through candles lose strength

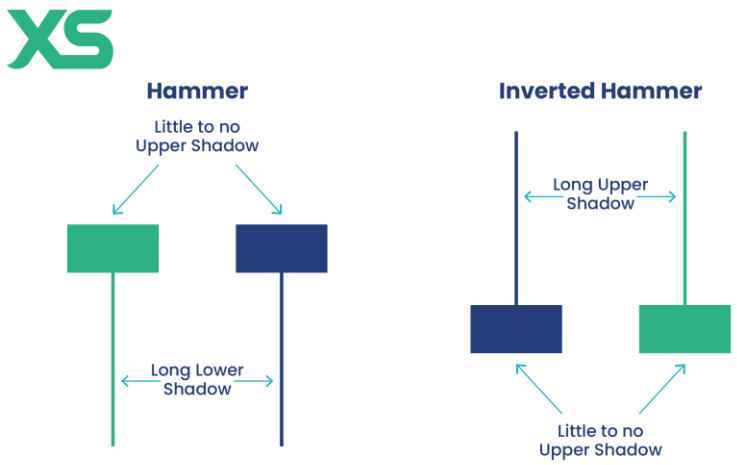

Figure 3.3.f. Hammer/ Inverted Hammer Candlestick Chart Pattern

g. Hanging Man / Inverted Hanging Man

- Hanging Man (Bearish): After uptrend, long lower wick, small body at top
- Inverted Hanging Man (Bearish): After uptrend, long upper wick
- Strategy:
    - ❖ Entry: Below low of the hanging man on red candle
    - ❖ Stop Loss: Above high of the hanging man
    - ❖ Target: Nearest swing low
    - ❖ Ext: If market turns neutral (Doji, low volume)

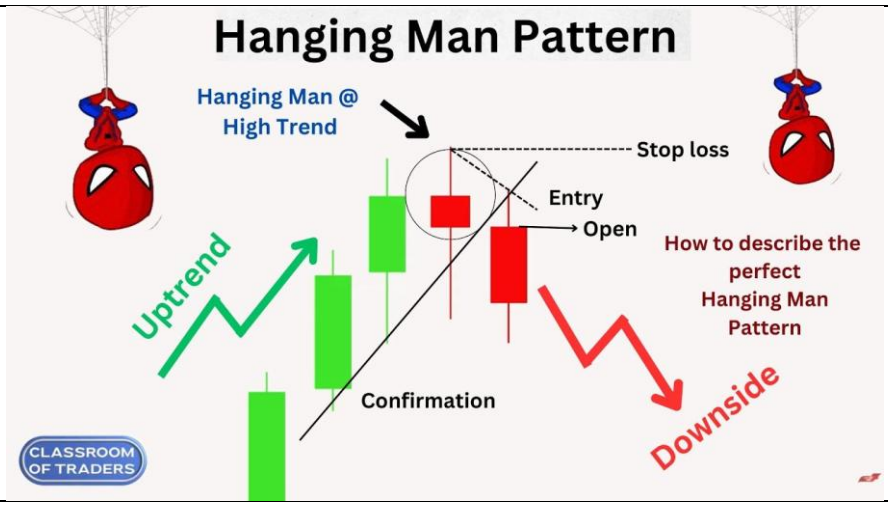

Figure 3.3.g. Hanging Man / Inverted Hanging Man Candlestick Chart Pattern

3.4. Gap Analysis in Candlestick Charts

Types of Gaps and Strategies are as follows:

- Common Gap: Random, usually in sideway trends; Avoid unless gap is filled with strong volume.
- Breakaway Gap: Appears at start of new trend.
    - ❖ Entry: After confirmation candle with volume
    - ❖ Stop Loss: Below gap low
    - ❖ Target: Measured move based on prior swing
- Runaway Gap: In middle of strong trend.
    - ❖ Entry: Add to position dulling pullback
    - ❖ Stop Loss: Below most recent swing
    - ❖ Target: Ride trend until exhaustion signal
- Exhaustion Gap: Final gap in a trend.
    - ❖ Entry: Opposite direction once reversal is confirmed
    - ❖ Stop Loss: Above/ below exhaustion wick
    - ❖ Target: Support/ resistance or Fibonacci levels

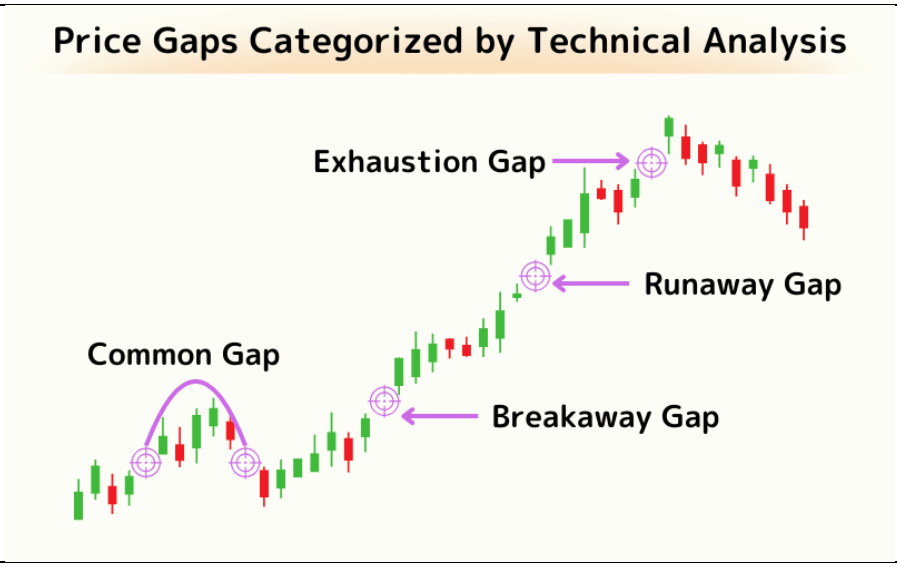

Figure 3.4. Gaps noted in Candlestick Charts

4. TECHNICAL ANALYSIS AND THEORIES IN ACTION

Technical Analysis is not just about indicators or patterns; it is underpinned by key market theories that help explain why price behaves the way it does. In this section, we look into five major theories taught during the session of Advanced Technical Analysis, and how they were applied or tested in actual market trades.

4.1. Dow Theory – Trend Structure and Market Phases

- Key Principles:
  - ❖ The market has three trends: Primary (Long-Term), Secondary (Correction), Minor (Short-term)
  - ❖ Prices move in Phases: Accumulation – Public Participation – Distribution
  - ❖ Trends are confirmed by volume
- Application:
  - ❖ Identified or accumulation phase In Nifty 50 sing higher lows plus an increase in the volume.
- How it helped:
  - ❖ Provided confidence to holder longer positions.
  - ❖ Helped distinguish between correction vs reversal.

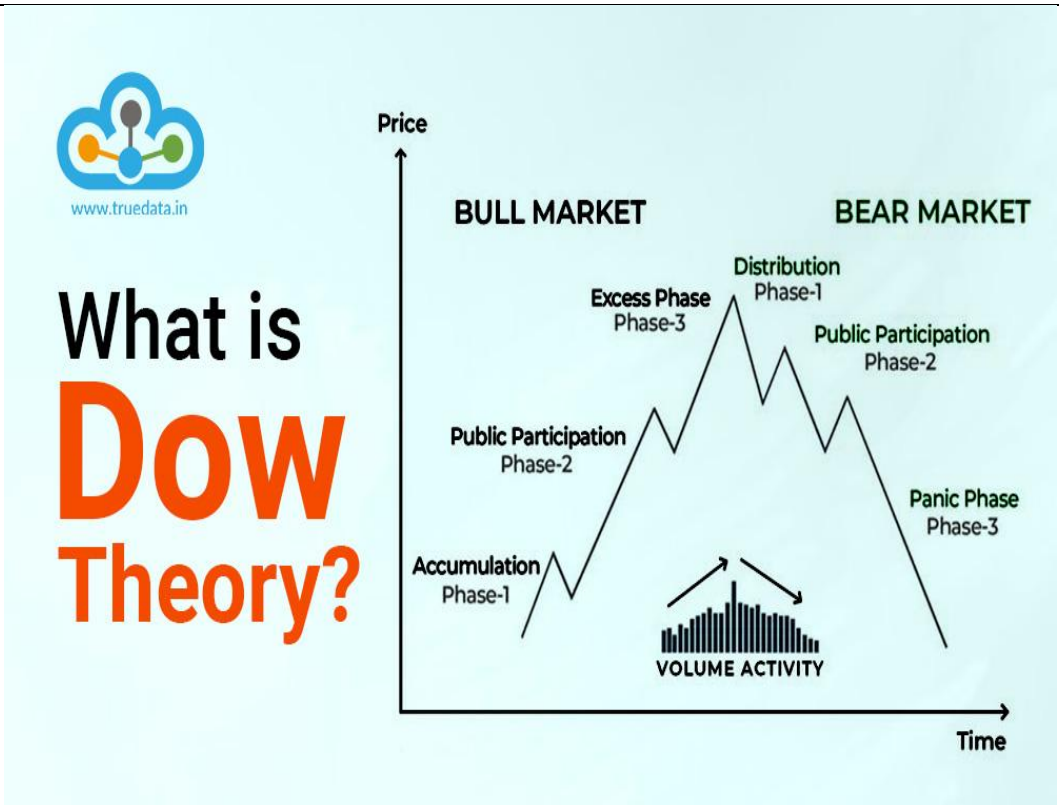

Figure 4.1. Dow Theory Market Phases

4.2. Eliott Wave Theory – Market Moves in Waves

- Key Concepts:
  - ❖ Five-wave pattern in trend (Impulse: 1,2,3,4,5) followed by a 3-wave correction (ABC).
  - ❖ Waves 3 and C are usually the strongest.
- Application:
  - ❖ Used wave structure on Bank Nifty to predict end of Wave 2 pullback and enter at the start of Wave 3.
  - ❖ Combined with Fibonacci retracement to identify entry at 61.8%.
- Challenges:
  - ❖ Wave counts are subjective and prone to misinterpretation in fast moving markets.
- Learning:
  - ❖ Useful for macro trend framing, not ideal for short-term entries without confirmation.

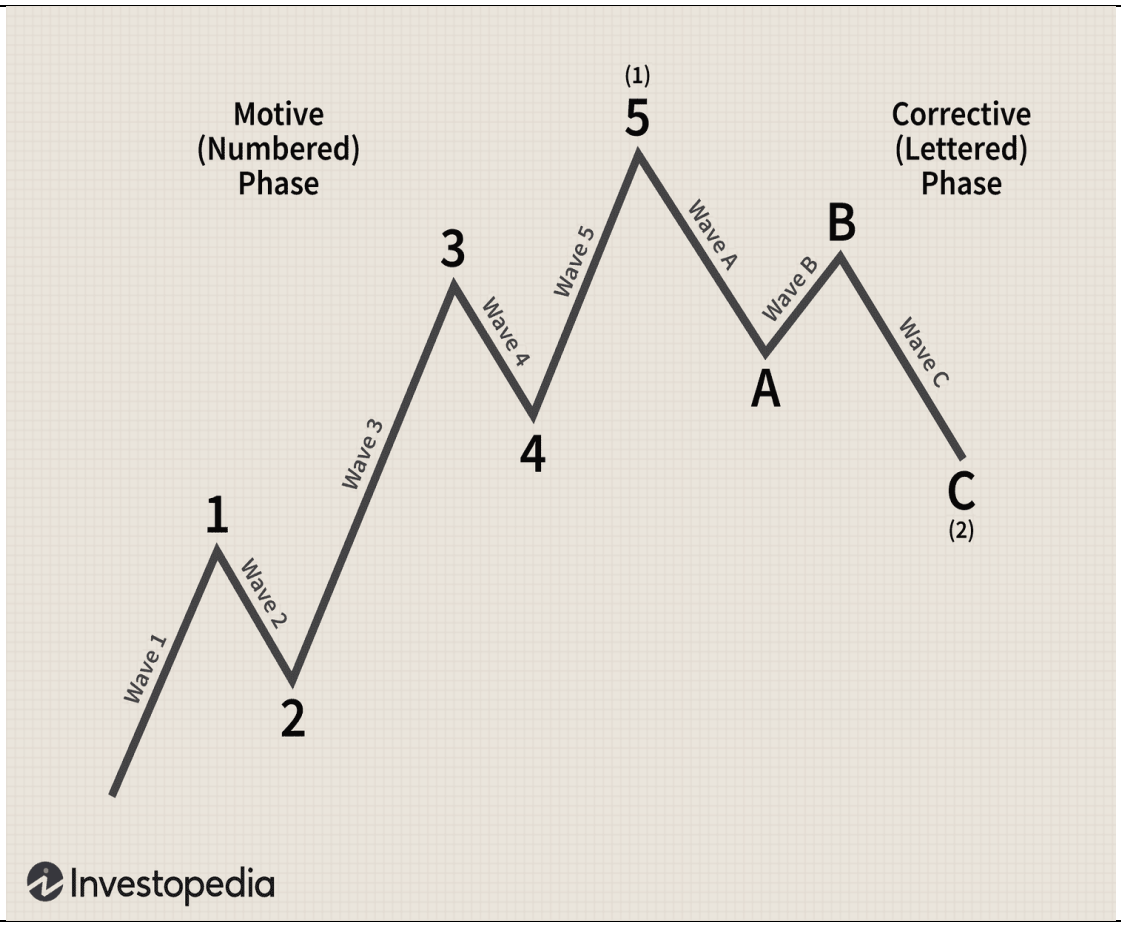

Figure 4.2. Eliott Wave Theory Phases

4.3. Fibonacci Retracement – Measuring Probable Pullbacks

- Key Levels:
  - 23.6%, 38.2%, 50%, 61.8%, 78.6%
- Application:
  - Entry tool: Bought HDFC Life at 61.8% retracement prior to up-move.
  - Ext Tool: Booked profits in Tata Motors near 161.8% extension of pervious swing.
  - Stop-Loss Placement: Just below the 78.6% level to allow volatility room.
- How it helped:
  - Provided a structured approach to planning entries and exits.
  - Reinforced belief in technical symmetry in price movement.

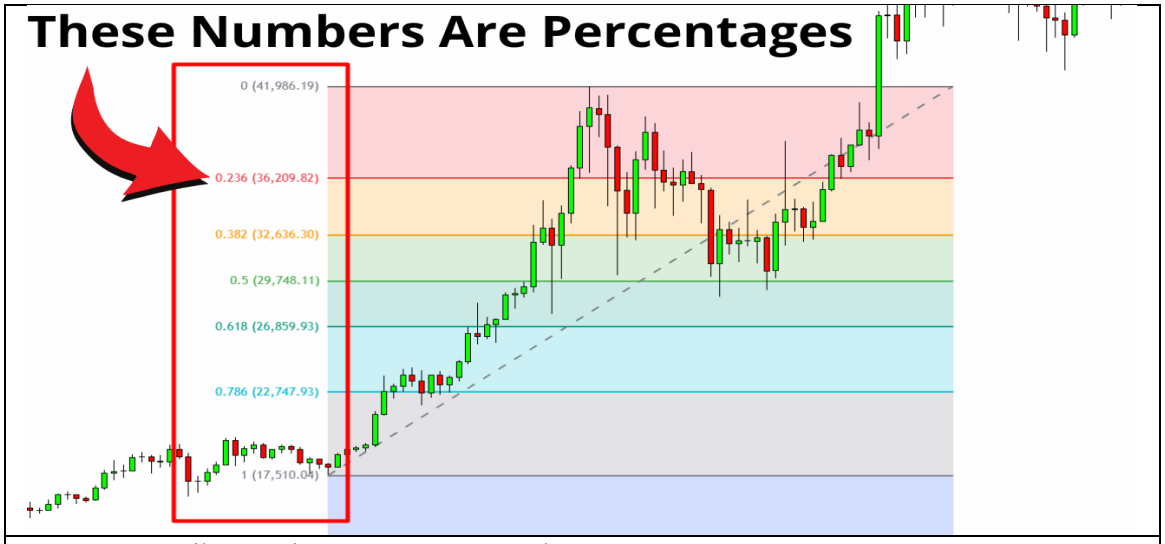

Figure 4.3. Fibonacci Retracement Levels

4.4. Efficient Market Hypothesis (EMH) – Challenge to Technical Analysis

- Theory:
  - All known information is already reflected in the price; no consistent alpha possible via charts.
- Contradiction:
  - Trades based on RSI divergence and Bollinger Band squeezes generated consistent short-term profits.
  - Gap trading strategies (especially breakaway gaps) yielded above-average returns.
  - EMH implies randomness, yet repeating pattern and reactions to indicators were clearly observed.
- Learning:
  - EMH may apply in large-cap stocks with high institutional coverage. (Eg: HUL, Infosys)
  - But technical inefficiencies still exist, especially in midcaps and during earnings/news events.

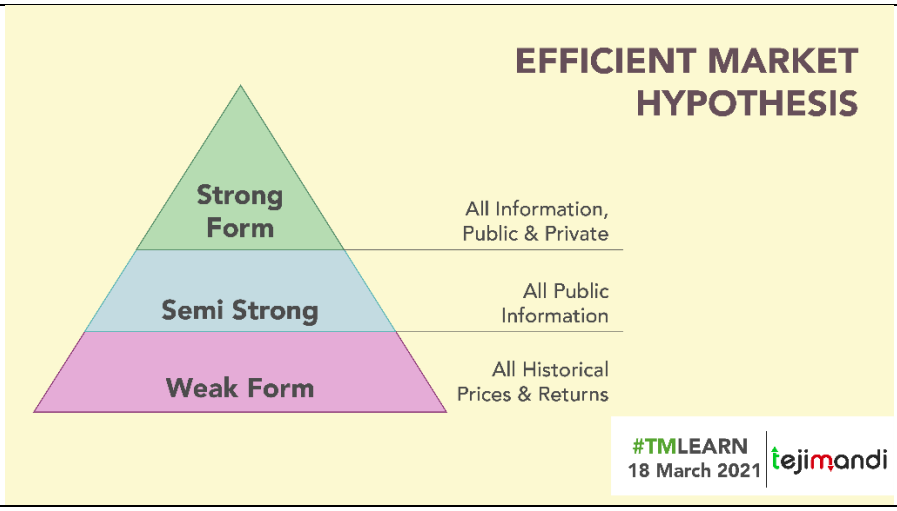

Figure 4.4. Efficient Market Hypothesis

4.5. Random Walk Theory – Price is Unpredictable

- Theory:
  - ❖ Price movements are random and past patterns have no predictive power.
  - ❖ Historical prices cannot be used to forecast future direction.
- Application & Refutation:
  - ❖ Flag, Triangle, and Cup Patterns worked repeatedly across stocks like IT, Infosys, and Bajaj Finance.
  - ❖ Fibonacci pullbacks and breakout retests showed statistically predictable behaviour.
- Learning:
  - ❖ Markets may appear random short-term, but behavioural tendencies and patterns emerge under volume pressure.
  - ❖ Random walk fails during earnings, events, and breakouts; clear structure emerges.

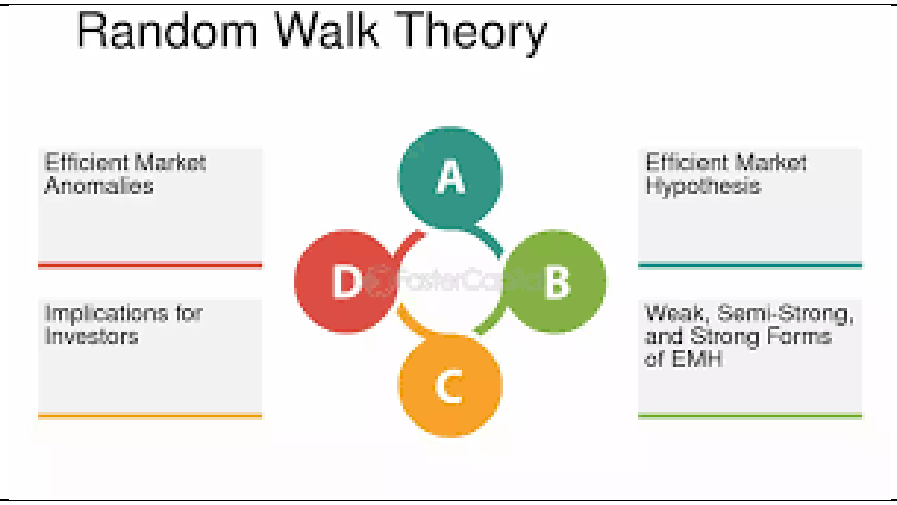

Figure 4.5. Random Walk Theory

5. DERIVATIVES MARKET INTEGRATION

Derivatives data provides an insight on a more institutional level regarding trader sentiment, trader positioning, and the expected volatility in the market. Incorporating derivatives data with your technical indicators and patterns increases the probability of success and confidence in your trades. This section will discuss the ways in which derivatives, and open interest, specifically, option chain, and Greeks (Delta, Gamma, Vega, Theta), were utilized in your trades using Nifty 50 and Nifty 200 equities.

5.1. Overview of Derivative Instruments Used

a. Futures: Directional positions in Nifty, Bank Nifty and selected stocks.
b. Options: Calls and Puts for intraday/intraweek directional bets and hedging.
c. Expiry Focus: Weekly options expiry used to identify gamma spikes and short-term volatility.
d. Tools Used: NSE Option Chain, TradingView F&O OI Data.

5.2. Open Interest (OI) Analysis

Why Open Interest matters?

- Open Interest shows whether new money is flowing into a contract, and in which direction. It is crucial to identifying whether the moves are speculative, confirmatory or exhaustive in nature.

Price Movement	OI Movement	Interpretation	Sentiment
Price ↑	OI ↑	Long Build-up	Bullish
Price ↓	OI ↑	Short Build-up	Bearish
Price ↑	OI ↓	Short Covering	Bullish Reversal
Price ↓	OI ↓	Long Unwinding	Bearish Reversal

Application in Trades:

- Short Covering in Nifty Futures confirms long entry after RSI bounce.
- Open Interest Analysis is used as secondary filter after pattern and indicator confirmation.

5.3. Option Chain Analysis: Support & Resistance Zones

a. Observed:
    - High Open Interest Call leads to a Resistance Zone.
    - High Open Interest Put leads to a Support Zone.
    - Shifting Open Interest Levels intraday indicates changing market sentiment.
b. Strategies that can be used:

- Look for Put writing increase at key Fibonacci levels, go for a long position accordingly.
- Call unwinding + Bulllish MACD = Confirmation of a positive breakout.

5.4. Option Greeks Used in Decision Making

Greek	Meaning	When It Helped
Delta	Sensitivity to price movement	Traded ITM options with stable delta near expiry
Gamma	Rate of change of Delta	Cautious during expiry days with high gamma moves
Theta	Time decay	Avoided buying options late in the week due to decay
Vega	Sensitivity to volatility	Bought options before events like RBI policy/US Fed

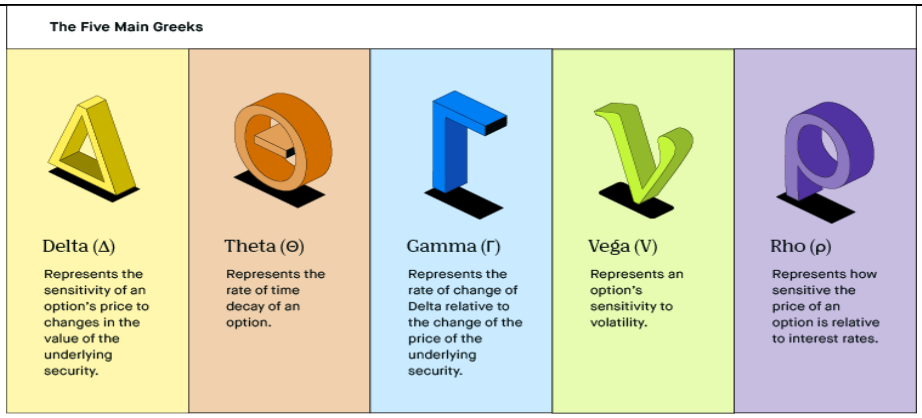

Figure 5.4. Option Greeks used in Options

5.5. Combining Technical with Derivatives for Trade Confirmation

Trade Setup	Technical Signal	Derivatives Confirmation	Action
Bullish Flag on Nifty	Price breakout + volume	Short covering in Futures + Call unwinding	Went long
Bearish Divergence on Infosys	RSI >70 + MACD divergence	Short build-up in Futures	Shorted
Morning Star in TCS	Candlestick reversal	Put writing at support + Vega rising	Bought Call Option
No pattern in Tata Steel	RSI neutral, sideways	Low OI build-up, no clear Greeks	Skipped trade

5.6. Nifty 50 and Nifty 200 Focus

- Trades were generally executed in liquid and option-active stocks.

- Nifty 50 used for directional trades and Bank Nifty used for volatility-based setups.
- Nifty 200 stocks such as Tata Chemicals, Canara Bank were only considered when accompanied by volume + OI
- Concentrate on stocks with tight spreads and higher option volumes.

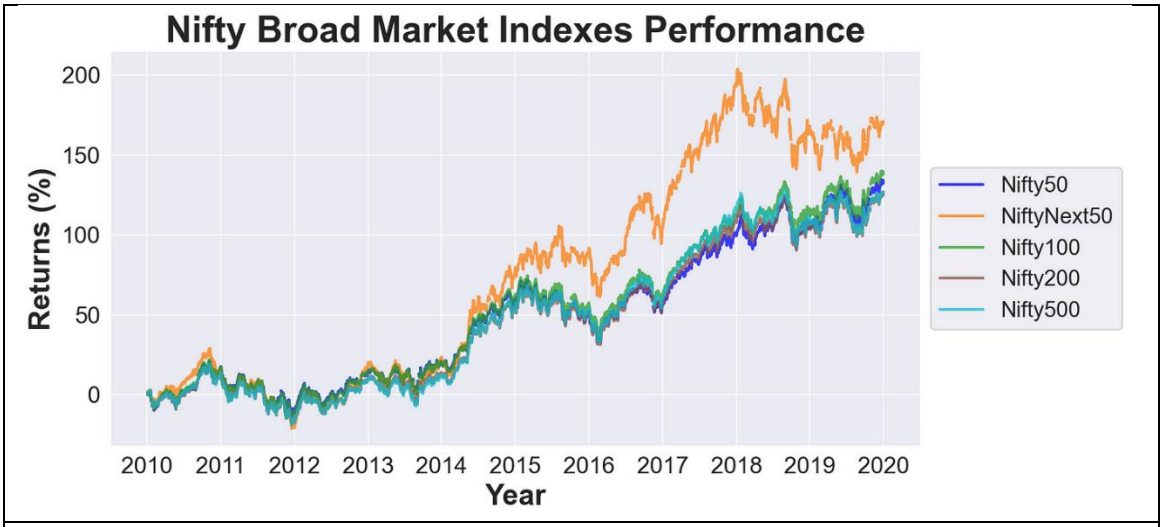

Figure 5.6. Nifty Major Indices Performances based on Returns

6. TRADE SUMMARY TABLE

This section provides a comprehensive summary of all trades executed during the duration of the capstone project. The trades listed below include both equity and derivative positions taken across Nifty 50 and Nifty 200 stocks, based on a structured approach involving technical indicators, chart patterns, and volume analysis.

Each trade entry captures essential details such as the stock name, entry and exit points, technical rationale, stop-loss and target levels, and the final outcome. This tabular representation is intended to offer a clear, concise record of trading activity, while the analysis, interpretation, and learnings from these trades are elaborated in top number 7 – Detailed Analysis of Key Trades.

Date	Sector	Company Name	Quantity	Entry	Exit	Stop Loss	Target	Profit and Loss	Estimated Charged	Net Profit and Loss
23-06-2025	Energy	ONGC	1000	251.45	251.20	251.20	251.85	-250.00	135.75	-385.75
23-06-2025	Energy	GAIL	1000	181.90	182.55	181.40	182.55	650.00	111.49	538.51
23-06-2025	Energy	GAIL	2000	182.10	182.50	181.70	182.50	800.00	175.80	624.20
23-06-2025	Energy	IOC	2000	137.44	137.90	137.10	137.90	920.00	144.34	775.66
24-06-2025	Energy	GAIL	1000	186.10	186.50	186.50	185.00	-400.00	112.84	-512.84
24-06-2025	Energy	ONGC	1000	246.80	245.70	247.48	245.70	1,100.00	134.12	965.88

Total Portfolio	1,00,00,000.00
Net P&L	2,005.66
Net Portfolio	1,00,02,005.66

Figure 6. Summary of the Key Trades and Realised Net Profit/ (Loss) from Front Page

A total of 6 trades were executed across four energy sector stocks; ONGC, GAIL, IOC, and GAIL, as on 23rd and 24th June 2025. The trades were predominantly intraday in nature and executed based on technical triggers such as support/resistance levels, candlestick patterns, and volume confirmation. While some trades achieved positive returns, others incurred minor losses due to early reversals or tight stop-loss levels. The net result across all trades was a modest overall profit of ₹2,005.66, reflecting the importance of precision and discipline even in short-term tactical strategies.

7. DETAILED ANALYSIS OF KEY TRADES

The following section provides an in-depth analysis of selected trades executed during the course of this capstone project. Each trade is dissected to illustrate how a confluence of technical indicators, chart patterns, volume behaviour, and derivatives data informed my decision-making process. The objective was not merely to profit from these trades, but to refine my ability to structure trades with a clear rationale, defined risk, and realistic targets.

For each trade, I have detailed the:

- Technical setup (e.g., chart pattern, indicator signals)
- Derivative insights (e.g., open interest trends, option chain data, Greeks)
- Volume and sentiment confirmation
- Entry, stop-loss, and exit strategies
- Actual outcomes and key learnings

These trades were selected to reflect a diversity of setups—ranging from breakout trades and pullback entries to gap trades and reversal patterns—across Nifty 50 and Nifty 200 stocks, with both winning and losing outcomes included for objective analysis.

a. Trade Number 1:

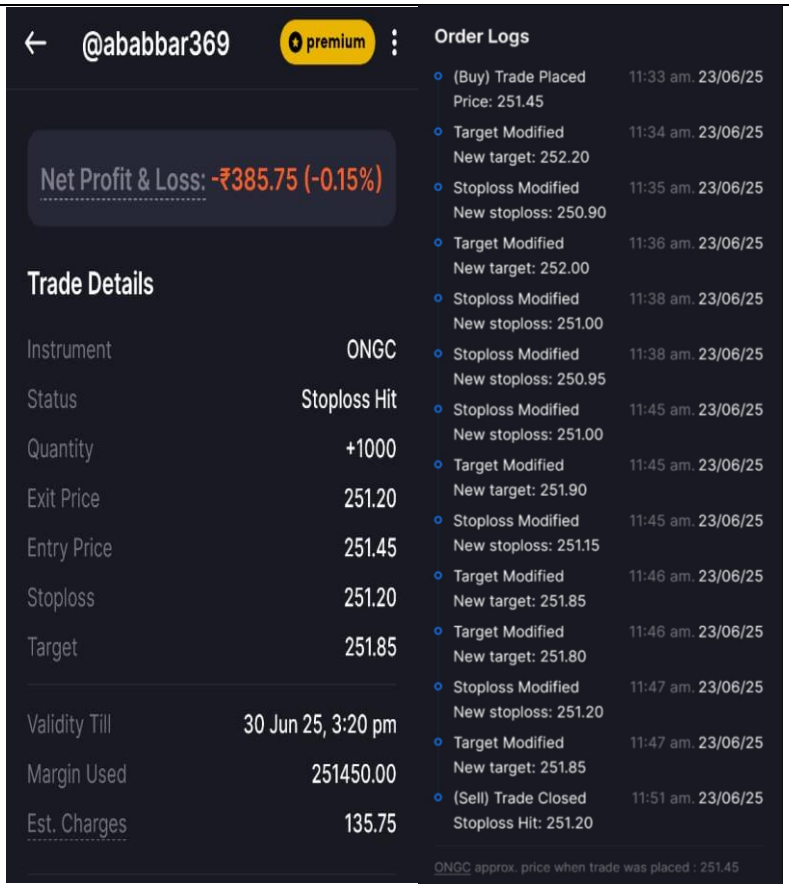

Figure 7.a. Snapshot of Technical Indicators & Trade Details used to execute Trade Number 1

- **Date:** 23/06/2025
- **Time Frame Used:** 5 minutes
- **Scrip:** ONGC
- **Entry Price:** 251.45
- **Exit Price:** 251.20
- **Stop Loss:** 251.20
- **Target:** 251.85
- **Net Profit and Loss:** -385.75
- **Reason for Entry/Exit:** The middle Bollinger became the support from resistance, but it was a false breakout and hence my stoploss was hit.
- **Indicators Used:** Use of Bollinger Bands and Net Volumes
- **Tools Used:** Fibonacci Retracements
- **What strategy is used:**
  Using a simple strategy using Bollinger bands, where if the price is rising towards the middle Bollinger band, and it breaks the middle Bollinger band, I will go long and set the upper middle band as the target, middle Bollinger band as the entry point, and lower Bollinger band as the stop loss. Likewise, if the Bollinger bands is headed towards the middle Bollinger band from the upper Bollinger band and break the middle Bollinger band, I will set the upper Bollinger band as the stop-loss, middle band as the entry price and the lower band as the target. I also incorporate Fibonacci retracement as a tool for double confirmation in terms of determining the resistance and support levels.
- **What did the option chains indicate in terms of Open Interest?**
  Strong call OI resistance seen around ₹245–250 strikes (>12 million OI), acting as overhead supply and capping upside.
- **Reflection Points:** It is crucial to wait for a double confirmation in terms of breakout, crucial to beware of false breakouts.

b. Trade Number 2:

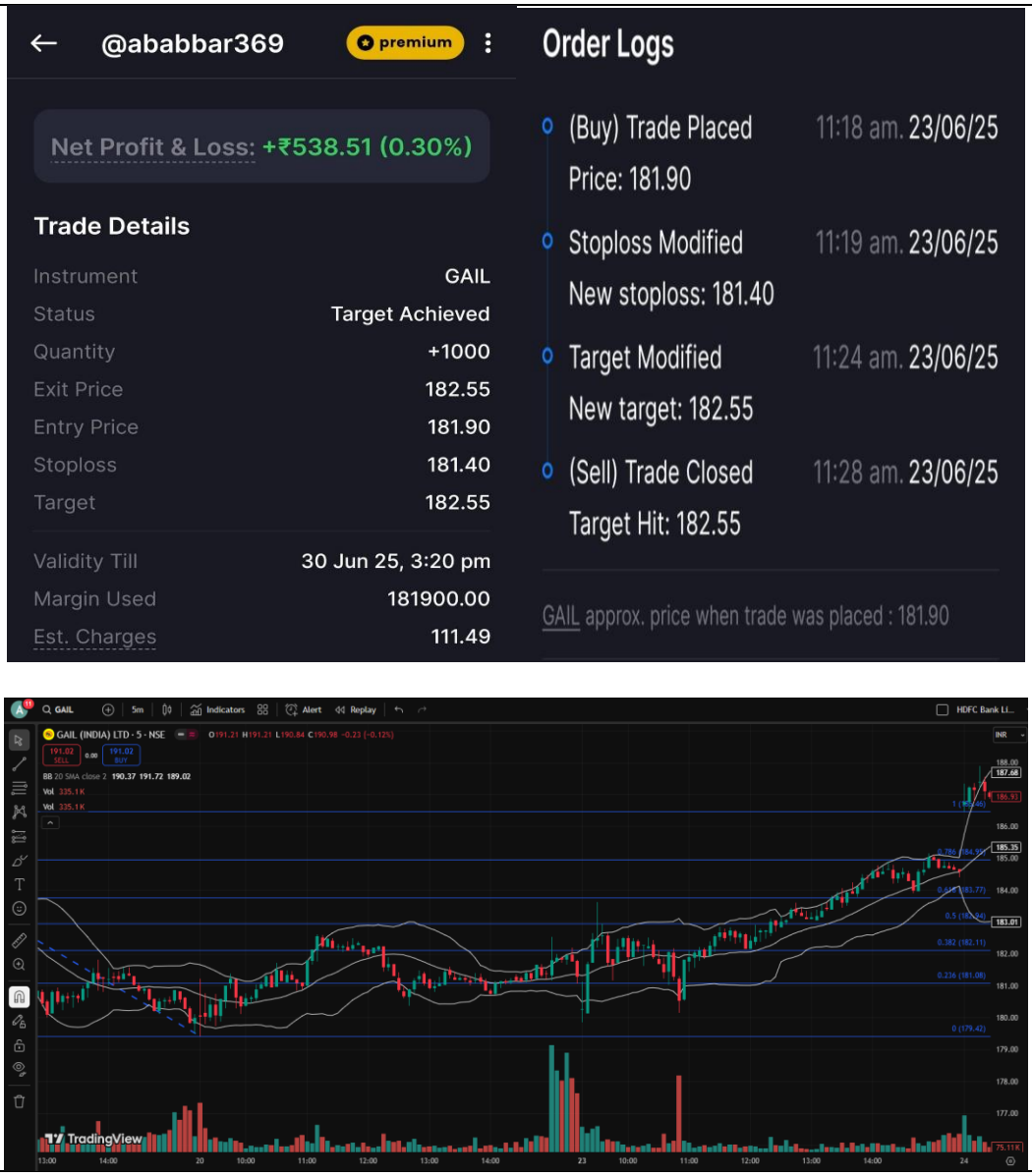

Figure 7.b. Snapshot of Technical Indicators & Trade Details used to execute Trade Number 2

- **Date:** 23/06/2025
- **Time Frame Used:** 5 Minutes Used
- **Scrip:** GAIL
- **Entry Price:** 181.90
- **Exit Price:** 182.55
- **Stop Loss:** 181.40
- **Target:** 182.55
- **Net Profit and Loss:** 538.51
- **Reason for Entry/Exit:** The middle Bollinger became the support from resistance, and my target was achieved.
- **Indicators Used**: Use of Bollinger Bands and Net Volumes

- **Tools Used:** Fibonacci Retracements
- **What strategy is used?**
  At 11:18 am, the price broke the middle band and hence I am using a simple strategy using Bollinger bands, where if the price is rising towards the middle Bollinger band, and it breaks the middle Bollinger band, I will go long and set the upper middle band as the target, middle Bollinger band as the entry point, and lower Bollinger band as the stop loss. Likewise, if the Bollinger bands is headed towards the middle Bollinger band from the upper Bollinger band and break the middle Bollinger band, I will set the upper Bollinger band as the stop-loss, middle band as the entry price and the lower band as the target. I also incorporate Fibonacci retracement as a tool for double confirmation in terms of determining the resistance and support levels.
- **What did the option chains indicate in terms of Open Interest?**
  Heavy put OI support at ₹180 strike (~1.47 million OI) provided a solid base for the bullish bounce.
- **Reflection:** When the price is rising from the below Bollinger band towards the middle Bollinger band, it is a great time to go long once the middle Bollinger band transitions from a support to a resistance.

c. **Trade Number 3:**

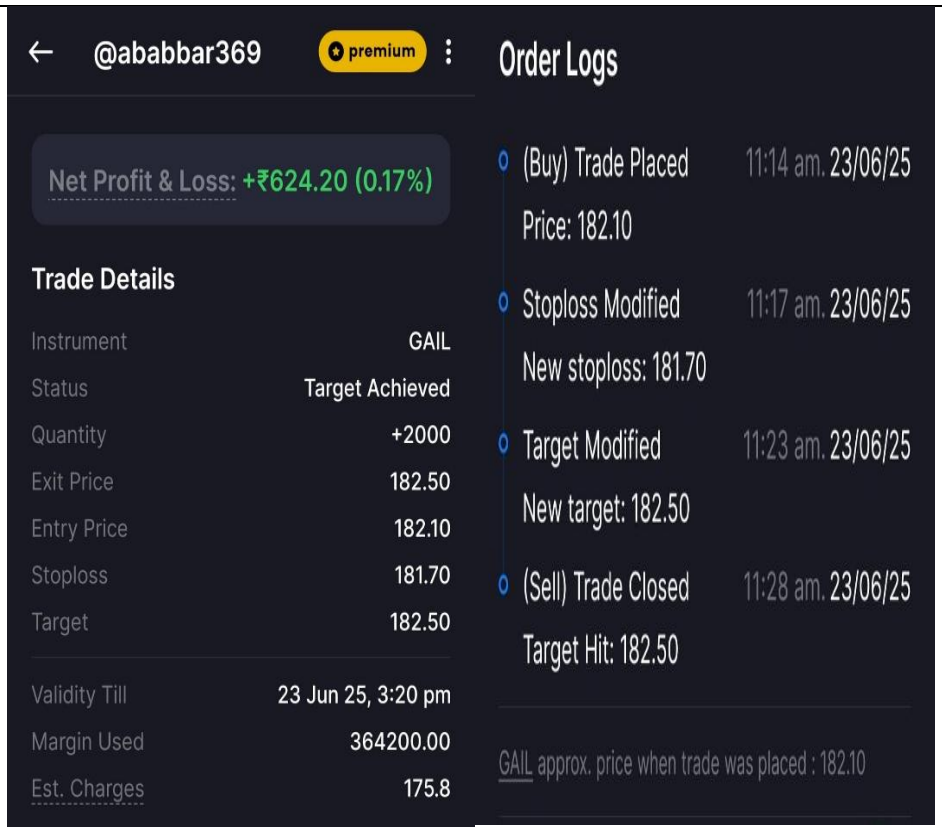

Figure 7.c. Snapshot of Technical Indicators & Trade Details used to execute Trade Number 3

- **Date:** 23/06/2025
- **Time Frame Used:** 5 Minutes Used
- **Scrip:** GAIL
- **Entry Price:** 182.10
- **Exit Price:** 182.50
- **Stop Loss:** 181.70
- **Target:** 182.50
- **Net Profit and Loss:** 624.20
- **Reason for Entry/Exit:** The middle Bollinger became the support from resistance, and my target was achieved.
- **Indicators Used:** Use of Bollinger Bands and Net Volumes
- **Tools Used:** Fibonacci Retracements
- **What strategy is used:**
  At 11:14 am, the middle Bollinger band became the support after the price broke it as the resistance, hence decide to go long by using a simple strategy using Bollinger bands, where if the price is rising towards the middle Bollinger band, and it breaks the middle Bollinger band, I will go long and set the upper middle band as the target, middle Bollinger band as the entry point, and lower Bollinger band as the stop loss. Likewise, if the Bollinger bands is headed towards the middle Bollinger band from the upper Bollinger band and break the middle Bollinger band, I will set the upper Bollinger band as the stop-loss, middle band as the entry price and the lower band as the target. I also incorporate Fibonacci retracement as a tool for double confirmation in terms of determining the resistance and support levels.
- **What did the option chains indicate in terms of Open Interest?**
  Continued put OI buildup at ₹180 confirmed strong support zone, reinforcing the long setup.
- **Reflection:** When the price is rising from the below Bollinger band towards the middle Bollinger band, it is a great time to go long once the middle Bollinger band transitions from a support to a resistance.

d. Trade Number 4:

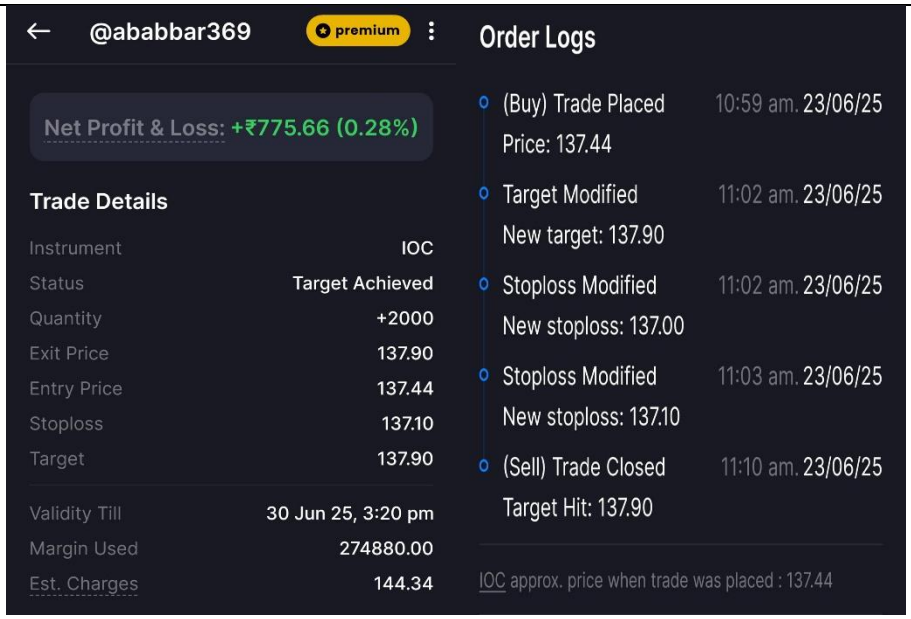

Figure 7.d. Snapshot of Technical Indicators & Trade Details used to execute Trade Number 4

- **Date:** 23/06/2025
- **Time Frame Used:** 5 Minutes Used
- **Scrip:** IOC
- **Entry Price:** 137.44
- **Exit Price:** 137.90
- **Stop Loss:** 137.10
- **Target:** 137.90
- **Net Profit and Loss:** 775.66
- **Reason for Entry/Exit:** The middle Bollinger became the support from resistance, and my target was achieved.
- **Indicators Used:** Use of Bollinger Bands and Net Volumes

- **Tools Used:** Fibonacci Retracements
- **What strategy is used?**
  At 10:59 am, the middle Bollinger band became the support from the resistance, and based on this, executing a simple strategy using Bollinger bands, where if the price is rising towards the middle Bollinger band, and it breaks the middle Bollinger band, I will go long and set the upper middle band as the target, middle Bollinger band as the entry point, and lower Bollinger band as the stop loss. Likewise, if the Bollinger bands is headed towards the middle Bollinger band from the upper Bollinger band and break the middle Bollinger band, I will set the upper Bollinger band as the stop-loss, middle band as the entry price and the lower band as the target. I also incorporate Fibonacci retracement as a tool for double confirmation in terms of determining the resistance and support levels.
- **What did the option chains indicate in terms of Open Interest?**
  Notable call OI cluster at ₹140–143 strikes (~3.4 million OI at ₹140 CE and 1.36 million at ₹143 CE) served as an intraday resistance zone, suggesting likely supply in that range.
- **Reflection:**
  When the price is rising from the below Bollinger band towards the middle Bollinger band, it is a great time to go long once the middle Bollinger band transitions from a support to a resistance.

e. Trade Number 5:

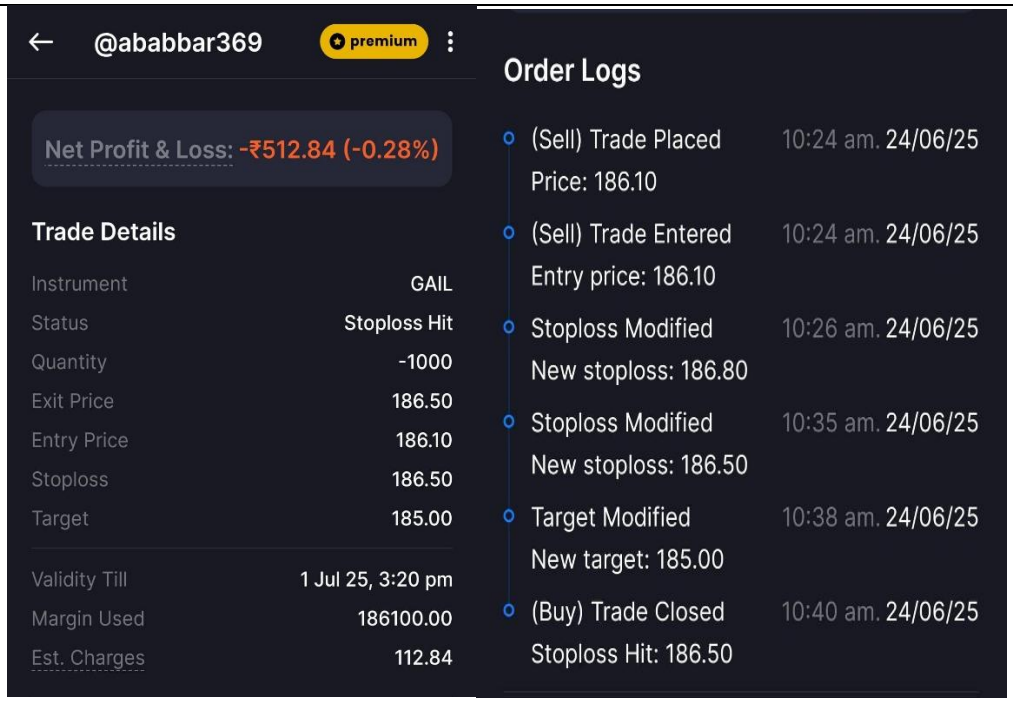

Figure 7.e. Snapshot of Technical Indicators & Trade Details used to execute Trade Number 5

- **Date:** 24/06/2025
- **Time Frame Used:** 5 minutes used
- **Scrip:** GAIL
- **Entry Price:** 186.10
- **Exit Price:** 186.50
- **Stop Loss:** 186.5
- **Target:** 185
- **Net Profit and Loss:** -512.84
- **Reason for Entry/Exit:** The Middle Bollinger became the resistance from support, but it was a false breakout and hence my stoploss was hit.
- Indicators Used: Use of Bollinger Bands and Volumes
- **Tools Used:** Fibonacci Retracements
- **What strategy is used:**
  At 10:14 am, it had reached it's high at 187.89 (Fibonacci 0 Retracement), and it was headed towards the middle Bollinger band, and it broke the middle band downwards, hence made an entry for short selling at 186.10, but as the volumes were low, it turned out to be a false breakout and used Fibonacci 23.6% (185.99) as the support and it did not break it, thereby I re-adjusted my Stop loss and exited at 186.50 as it bounced back for an up move.
- **What did the option chains indicate in terms of Open Interest?**
  Significant call OI accumulation at ₹190 (~5.9 million OI) created resistance, validating the short entry.
- **Reflection:** It is crucial to wait for a double confirmation in terms of breakout, crucial to beware of false breakouts.

f. Trade Number 6:

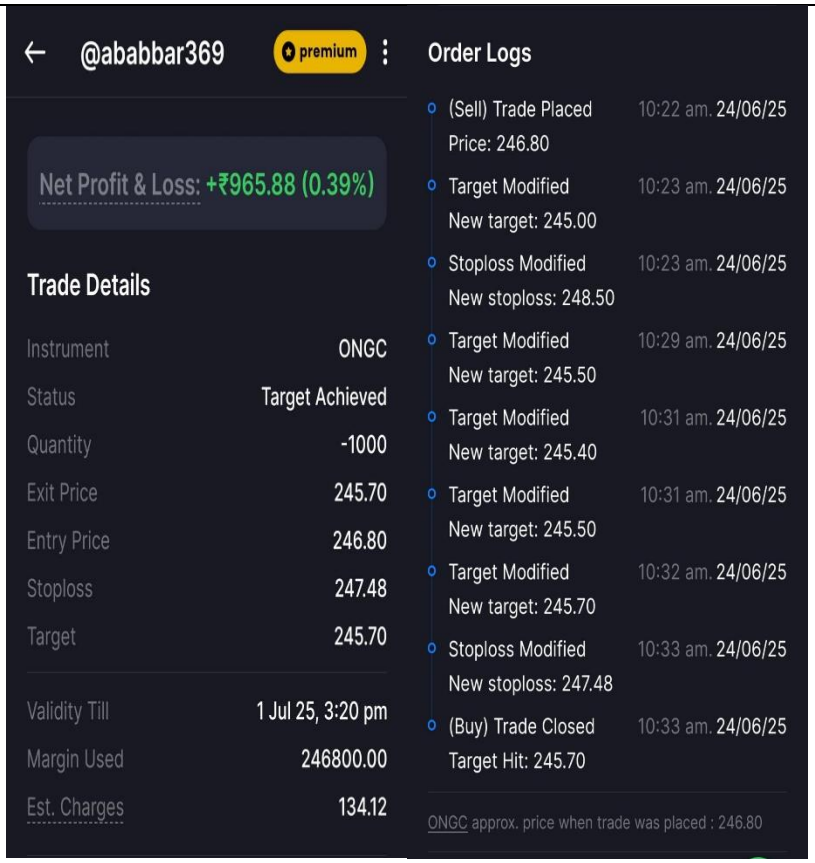

Figure 7.f. Snapshot of Technical Indicators & Trade Details used to execute Trade Number 6

- **Date:** 24/06/2025
- **Time Frame Used:** 5 minutes used
- **Scrip:** ONGC
- **Entry Price:** 246.8

- **Exit Price:** 245.70
- **Stop Loss:** 247.48
- **Target:** 245.70
- **Net Profit and Loss:** 965.88
- **Reason for Entry/Exit:** The middle Bollinger became the resistance from support, and my target was achieved.
- **Indicators Used:** Use of Bollinger Bands and Net Volumes
- **Tools Used:** Fibonacci Retracement
- **What strategy is used?**
  At 10:22 am, the price moved from the upper Bollinger and broke the middle Bollinger band, thereby ended up using the Using a simple strategy using Bollinger bands, where if the price is rising towards the middle Bollinger band, and it breaks the middle Bollinger band, I will go long and set the upper middle band as the target, middle Bollinger band as the entry point, and lower Bollinger band as the stop loss. Likewise, if the Bollinger bands is headed towards the middle Bollinger band from the upper Bollinger band and break the middle Bollinger band, I will set the upper Bollinger band as the stop-loss, middle band as the entry price and the lower band as the target. I also incorporate Fibonacci retracement as a tool for double confirmation in terms of determining the resistance and support levels.
- **What did the option chains indicate in terms of Open Interest?**
  Elevated call OI at ₹250–255 (~8–12 million), acting as a ceiling and supporting the bearish setup.
- **Reflection:**
  When the price is dipping from the Upper Bollinger band towards the middle Bollinger band, it is a great time to short once the middle Bollinger band transitions from a support to a resistance.

8. REFLECTION REPORT

This capstone project has been an eye-opening experience that bridged the gap between theoretical technical analysis frameworks and their practical applicability in live trading environments. Over the course of analysing and executing trades, I have gained technical proficiency as well as greater market intuition, risk, discipline, and an appreciation for the nuances of market psychology.

A. Application of Indicators and Tools

A key learning from this project was how to go beyond using indicators in isolation. Tools like Bollinger Bands, Fibonacci Retracement, and Volume analysis became crucial in terms of decision-making.

- Bollinger Bands:

I learned how to use the upper, middle and lower bands as a dynamic resistance and support levels. For long positions, I entered when prices approached the lower band with volume divergence, and booked profits near the middle or upper band. I also used the upper band for short positions, particularly when the candle patterns (Eg: Shooting Star or Bearish Engulfing) appeared near resistance with shrinking volume. This improved my entry timing and exit planning, reducing emotional exits and enhancing conviction.

- Fibonacci Retracement:

Rather than using pivot points for the support/resistance levels, Fibonacci levels of retracement allowed me to quantify potential pullbacks. Entry decision at the 61.8% retracement level, combined with pattern confirmations, gave me greater risk-reward setups. It was also effective in target setting using its extensions.

- Volume:

Volume is the true validator of any technical signal. A breakout without volume often resulted in a false signal, while a surge in volume during a pattern confirmation gave me the confidence to size up positions. It also helped me avoid low volume traps.

B. Pattern Recognition and Strategic Alignment

Understanding patterns such as Double Tops, Head & Shoulders, Flags, and Triangles became a core skill, but learning how to use them strategically was the real takeaway.

- I used patterns not just for entry signals but also to define stop-loss and target zones.
- In some cases, the market contradicted textbook expectations, helping me learn that pattern failure is also a powerful signal, especially when paired with derivatives data.

The discipline of always identifying the entry, stop-loss, and target before entering a trade made my approach more structured and professional.

C. Theories tested in the Market

I applied theories such as Dow Theory, Fibonacci, and Elliott Waves, while critically evaluating Efficient Market Hypothesis (EMH) and Random Walk Theory.

- I found that EMH didn't hold true in the short term, as repeating price behaviours, support/resistance reactions, and volume-backed breakouts clearly offered predictable patterns.
- However, in highly liquid large-cap stocks like ONGC or GAIL, some randomness and efficiency did exist, showing partial validation of the theory.

These reflections helped me understand that while theories provide structure, flexibility and contextual interpretation are essential for practical trading.

D. Psychological and Behavioural Learnings

Perhaps, the most important learning for me on a personal note was behavioural, as I learned to:

- Stick to my stop-loss even when the trade felt "almost right".
- Avoid overtrading, especially after losses.
- Maintain objectivity by relying on data rather than gut feeling.
- Accept that not all trades will work, and a good trade isn't always a profitable one; it's the one with correct logic, structure, and discipline, thereby sticking to one's process and strategy.

E. Integration of Derivatives Data

Integrating Open Interest, Option Chain analysis, and Greeks added a whole new layer to my decision-making. I learned that:

- Short covering and OI build-ups can validate chart-based long or short setups.
- Using option chain support/resistance zones helped me avoid trades that looked good on charts but had strong derivative resistance.
- Avoiding theta decay in options trading near expiry saved me from overpaying for time.

This allowed me to trade more in line with institutional flows, rather than retail sentiment.

F. Realization and Improvements

- I learned that having a technical setup is necessary but not sufficient as volume and sentiment are what make it tradable.
- I began to look at confluence zones—e.g., a 61.8% retracement + RSI bounce + bullish engulfing + short covering = high probability.

- I improved my trade journaling, which helped me review mistakes and avoid repeating them.

G. Conclusion

This project has transformed how I view the market in terms of technical analysis – from a static toolkit to a dynamic decision-making system. By integrating indicators, patterns, volume, and derivatives data, I have built a more structured and strategic trading approach.

As I continue my journey in the financial markets, these learnings will serve as the foundation for both intraday precision and long-term portfolio decisions. More than anything, this project taught me the importance of discipline, patience, and structured thinking in terms of trading, skills that are just as vital in investment banking, equity research, or portfolio management.